

AI-Based Credit Card Fraud Detection System

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Abstract

Credit card fraud has become a major challenge due to the rapid growth of digital payment systems and online transactions. Traditional fraud detection methods are often ineffective in identifying new and complex fraudulent activities, especially when dealing with highly imbalanced datasets. This study proposes a machine learning-based approach for detecting fraudulent credit card transactions using supervised classification techniques.

The proposed system applies three machine learning algorithms: Logistic Regression, Decision Tree, and Random Forest. To address the issue of class imbalance, the SMOTE (Synthetic Minority Over-sampling Technique) method was applied during data preprocessing. The models were trained and evaluated using the Credit Card Fraud Detection dataset obtained from Kaggle. Performance evaluation was conducted using Accuracy, Precision, Recall, F1-score, and ROC curve analysis.

The experimental results demonstrated that the Random Forest model achieved the best overall performance among all evaluated models, obtaining 99.95% accuracy, 92.41% precision, 76.84% recall, an F1-score of 83.91%, and an AUROC of 0.9519. In addition, the model effectively reduced false positive predictions while maintaining strong fraud detection capability. Nevertheless, approximately 23% of fraudulent transactions remained undetected, highlighting the ongoing challenge of maximizing fraud detection performance while minimizing false predictions in real-world financial systems. The findings confirm that machine learning techniques can significantly improve fraud detection performance and enhance financial transaction security.

1. Introduction

The rapid expansion of digital payment systems and online transactions has significantly increased the use of credit cards worldwide. While this advancement offers convenience and efficiency, it has also led to a rise in fraudulent activities. Credit card fraud poses a serious threat to both financial institutions and customers, resulting in substantial financial losses and reduced trust in digital systems [4][6].

Detecting fraudulent transactions is a challenging task due to the large volume of daily transactions and the complex patterns of fraud. Traditional rule-based detection systems are often insufficient in identifying new and evolving fraud techniques [4]. Therefore, there is a need for intelligent and automated systems that can effectively detect fraudulent activities.

Machine Learning (ML) has emerged as a powerful solution for fraud detection, as it can analyze large datasets and identify hidden patterns. Previous studies have demonstrated the effectiveness of machine learning techniques such as Random Forest and Deep Learning in fraud detection tasks [5][8]. These approaches have shown promising results in identifying complex fraud patterns and improving detection accuracy compared with traditional rule-based systems. However, many existing studies focus on individual models without providing a consistent comparison between multiple algorithms under the same experimental conditions.

In addition, fraud detection datasets are highly imbalanced, where fraudulent transactions represent only a small portion of the data. This imbalance can negatively affect model performance and requires appropriate techniques such as resampling methods like SMOTE to improve detection accuracy [2].

Furthermore, real-world fraud detection systems must balance fraud detection capability with operational efficiency. Excessive false positive predictions may incorrectly flag legitimate customer transactions and reduce user satisfaction, while missed fraudulent transactions can result in significant financial losses. Therefore, an effective fraud detection model should achieve a balance between identifying fraudulent activities and minimizing unnecessary alerts.

The aim of this study is to develop and evaluate multiple supervised machine learning models for detecting fraudulent credit card transactions and to identify the most effective model. The proposed approach includes data preprocessing, handling class imbalance using SMOTE, and applying classification algorithms such as Logistic Regression, Decision Tree, and Random Forest to identify the most effective fraud detection model.

The remaining part of this paper is organized as follows. Section 2 presents the related literature review. Section 3 describes the proposed machine learning techniques. Section 4 discusses the empirical studies, including dataset description and experimental setup. Section 5 presents the results and discussion, and finally, Section 6 concludes the study.

2. Review of Related Literatures

Suresh Kumar et al. [4] investigated the critical issue of online credit card fraud, a growing concern for financial institutions. Their study implemented a supervised learning framework using the Random Forest algorithm to enhance classification stability. By applying this model to a real-world transaction dataset, they aimed to distinguish legitimate activities from fraudulent ones. Using a confusion matrix for evaluation, the results demonstrated approximately 90% accuracy, confirming Random Forest as an effective approach for fraud detection. However, the study mainly focused on accuracy and did not consider issues such as class imbalance or model comparison.

In a similar context, Nguyen, Tran, and Nguyen [5] explored the potential of Deep Learning techniques for fraud detection. They conducted a comparative analysis between traditional machine learning algorithms and deep learning models across multiple datasets. Their findings indicated that deep learning outperformed classical approaches, especially in capturing complex and non-linear fraud patterns. Despite this, the study paid limited attention to feature engineering and computational efficiency.

Ileberi et al. [6] proposed a hybrid system combining multiple classification algorithms such as Decision Trees, Artificial Neural Networks, and Naive Bayes. A key contribution of their work was the use of a Genetic Algorithm for feature selection, which improved model performance by selecting the most relevant features. Although their approach improved accuracy and handled feature selection effectively, it did not adequately address class imbalance issues.

Alarfaj et al. [7] introduced a deep learning-based fraud detection system using Convolutional Neural Networks (CNN). Their study compared CNN with other machine learning algorithms such as XGBoost and Decision Trees. The results showed high accuracy and improved fraud detection performance, particularly under imbalanced conditions. However, the computational complexity of deep learning models was not fully discussed.

Zhichao Xie et al. [8] proposed a hybrid method that combines Mahalanobis Distance-based SMOTE-ENN with Random Forest. Their approach effectively handled extreme class imbalance and improved model performance across multiple evaluation metrics such as accuracy and AUC. In addition, the study incorporated feature relationships, making it more robust compared to earlier works.

Finally, Owen et al. [9] developed a multi-stage hybrid model that integrates K-Means clustering, Isolation Forest, and K-Nearest Neighbors (K-NN). Their model first clusters the data, then detects anomalies, and finally performs classification. This approach achieved a high accuracy of 99.68% and a recall of 100%, outperforming many standalone models. Despite its strong performance, the model introduces higher computational complexity, which may limit its real-time applicability.

Several previous studies, including Owen et al. [9] and Niu et al. [15], utilized the same Kaggle Credit Card Fraud Detection dataset adopted in this study. Their findings demonstrated the effectiveness of machine learning and ensemble learning techniques in detecting fraudulent transactions and serve as important benchmarks for evaluating and comparing the performance of the proposed models in this research.

Ref	Authors	Dataset	Method	Evaluation Metrics	Results
[4]	Suresh Kumar et al. (2019)	Credit card transaction dataset	Random forest (supervised learning)	Accuracy, Confusion Matrix	Achieved approximately 90% accuracy , demonstrating the effectiveness of Random Forest.
[5]	Nguyen, Tran, and Nguyen (2020)	European card datasets (ECD, SCD, TCD)	Deep Learning vs Traditional ML	Accuracy, Recall, F1-score	Deep learning models outperformed traditional machine learning methods, particularly in capturing complex fraud patterns.
[6]	Ileberi et al. (2022)	European credit card dataset	Genetic Algorithm + DT, RF, LR, ANN, NB	Accuracy, Precision, Recall, F1-score, AUC	The use of Genetic Algorithm for feature selection improved model performance and enhanced fraud detection accuracy.
[7]	Alarfaj et al. (2022)	European cardholders dataset	CNN, DT, RN, XG Boost	Accuracy, Precision, Recall, F1-score, AUC	Achieved up to 99.9% accuracy while reducing false negatives, indicating strong detection capability.
[8]	Zhichao Xie et al. (2024)	European and Taiwan datasets	SMOTE-ENN + Random Forest	Accuracy, Precision, Recall, F1-score, MCC, AUC	Successfully handled extreme class imbalance and improved detection performance across all evaluation metrics.

[9]	Owen et al. (2025)	Kaggle European Dataset	K-Means + Isolation Forest + K-NN	Accuracy, Precision, Recall, F1-Score	Achieved 99.68% accuracy and 100% recall , significantly outperforming standalone models.
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Table 1: Summary of related studies on credit card fraud detection.

Research Gap	Suresh Kumar et al. (2019)	Nguyen, Tran, and Nguyen (2020)	Ileberi et al. (2022)	Alarfaj et al. (2022)	Zhichao Xie et al. (2024)	Owen et al. (2025)
Model Comparison	✗	✓	✗	✓	✓	✓
Focus only on accuracy	✓	✗	✓	✓	✗	✗
Class Imbalance	✗	✓	✗	✓	✓	✓
Feature Engineering	✗	✓	✗	✗	✓	✓
Computational Complexity	✗	✗	✓	✗	✓	✓

Table 2: Summary of Research Gaps in Previous Studies

Gap Analysis

Despite the advancements in credit card fraud detection, several research gaps persist across existing studies. Early research, such as **Suresh Kumar et al. (2019)**, primarily focused on basic model implementation and accuracy, neglecting critical factors like model comparison and class imbalance. While more recent studies like **Nguyen, Tran, and Nguyen (2020)** successfully addressed class imbalance and model comparison, they often overlooked the importance of feature engineering or computational complexity. Furthermore, although **Zhichao Xie (2024)** and **Owen et al. (2025)** provided comprehensive frameworks that cover most technical gaps—including feature correlation and hybrid sampling, there is still a need for a robust fraud detection system that balances high recall with low computational cost, while effectively handling extreme class imbalance in real-world environments.

Therefore, this study proposes a machine learning-based fraud detection framework that combines class imbalance handling using SMOTE with a comparative evaluation of Logistic Regression, Decision Tree, and Random Forest classifiers. By evaluating multiple models under the same experimental conditions, the proposed approach aims to identify a solution that achieves strong fraud detection performance while maintaining reasonable computational efficiency and minimizing false predictions.

This section shows the deliverables of the project:

Deliverable	To whom	Delivery Media	Duration	Date
Project Proposal	Dr. Sarah Alyami	Softcopy	1 week	April 2, 2026
Final Project Report	Dr. Sarah Alyami	Softcopy	2 weeks	June 2, 2026
Final Project Presentation	Dr. Sarah Alyami	Softcopy	3 weeks	June 2, 2026

Table 3: Project Deliverables

3.0 Description of the Proposed Techniques

This project utilizes supervised machine learning techniques to detect fraudulent credit card transactions in a highly imbalanced dataset. The selected algorithms are widely used in classification problems and provide a balance between simplicity, interpretability, and performance.

Three classification models are implemented and compared: Logistic Regression, Decision Tree, and Random Forest. These models were selected to represent different levels of complexity, ranging from simple linear models to more advanced ensemble-based techniques. This enables a comprehensive comparison of their effectiveness in fraud detection. [10]

In addition, these models were selected because they offer different learning characteristics and levels of complexity. Logistic Regression serves as a simple and interpretable baseline model, Decision Tree provides interpretable decision rules and can capture non-linear patterns, while Random Forest offers improved robustness and predictive performance through ensemble learning. Comparing these models allows a comprehensive evaluation of different machine learning approaches for fraud detection under the same experimental conditions.

3.1 Logistic Regression

Logistic Regression is a statistical classification algorithm used to predict binary outcomes. In this project, it is applied to classify transactions as either legitimate or fraudulent.

Logistic Regression was selected as a baseline classifier due to its simplicity, computational efficiency, and ease of interpretation. It provides a useful benchmark for evaluating the effectiveness of more advanced machine learning models.

The model estimates the probability that a given transaction belongs to a particular class using a logistic (sigmoid) function. It is simple, efficient, and performs well when the relationship between input features and the target variable is approximately linear.

However, Logistic Regression has limitations in capturing complex, non-linear relationships in fraud data, which is a limitation in this project due to the complexity of fraudulent behaviors. [11]

3.2 Decision Tree

Decision Tree is a supervised learning algorithm that partitions the dataset into subsets based on feature values. It creates a tree-like structure where each node represents a decision rule.

Decision Tree was selected because it can model complex decision boundaries and provide interpretable classification rules. Its ability to handle non-linear relationships makes it suitable for fraud detection problems where transaction patterns may be difficult to separate using linear models.

This technique is easy to interpret and can handle both linear and non-linear relationships. It is capable of capturing complex decision boundaries and is well-suited for fraud detection tasks.

However, Decision Trees can suffer from overfitting, especially when the tree becomes too deep, which may affect the model's ability to generalize to unseen data. [12]

3.3 Random Forest

Random Forest is an ensemble learning method that combines multiple Decision Trees to improve performance and reduce overfitting.

Random Forest was selected as the primary candidate model because ensemble learning techniques have consistently demonstrated strong performance in fraud detection tasks. By combining multiple decision trees, the model can improve classification accuracy, reduce overfitting, and provide better generalization on unseen transactions.

Each tree in the forest is trained on a random subset of the data and features, and the final prediction is determined by aggregating the outputs of all trees. This approach enhances model stability and accuracy.

Random Forest is highly effective in handling complex datasets and generally provides better generalization than a single Decision Tree. It is also particularly suitable for imbalanced datasets, as it improves generalization and reduces bias toward the majority class.

However, Random Forest requires more computational resources and is less interpretable compared to simpler models. [13]

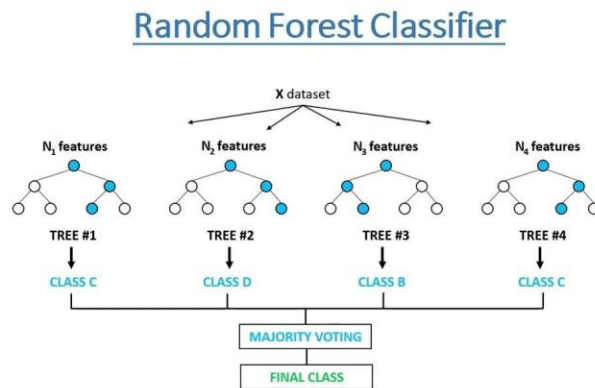


Figure 1. Structure of the Random Forest classifier showing multiple decision trees and aggregated prediction.

4.0 Empirical Studies

4.1 Description of Dataset

The dataset used in this study is the Credit Card Fraud Detection dataset obtained from the Kaggle repository [1]. It contains real-world credit card transactions made by European cardholders over a period of two days in September 2013.

This study addresses a binary classification problem, where the objective is to distinguish between legitimate and fraudulent transactions.

The dataset consists of 284,807 transactions and includes 30 features. Most of the features are anonymized numerical variables labeled from V1 to V28. The only original features that remain unchanged are Time and Amount. The target variable is Class, where 0 represents a legitimate transaction and 1 represents a fraudulent transaction.

Property	Value
Total Transactions	284,807
Fraudulent Transactions	492
Legitimate Transactions	284,315
Number of Features	30
Target Variable	Class
Fraud Percentage	0.17%

Table 4: Summary of the Credit Card Fraud Detection Dataset

One of the major challenges of this dataset is the severe class imbalance. Out of the total transactions, only 492 are fraudulent, representing approximately 0.17% of the dataset, while the remaining transactions are legitimate.

Figure 2 illustrates the distribution of legitimate and fraudulent transactions.

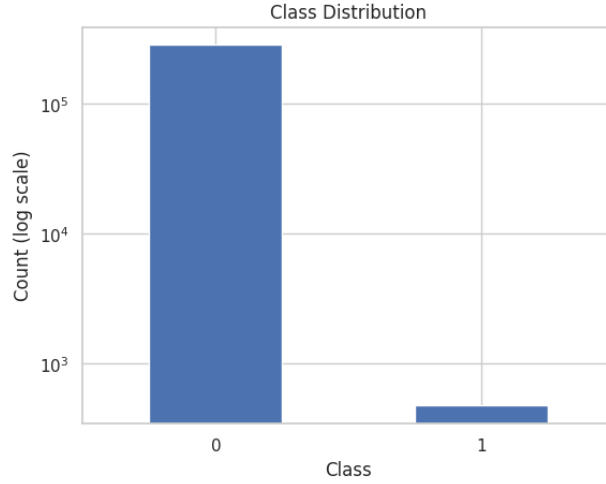


Figure 2: Class distribution of the dataset showing severe imbalance between legitimate (Class 0) and fraudulent (Class 1) transactions (log scale).

This extreme imbalance makes fraud detection a challenging task, as machine learning models tend to be biased toward the majority class.

Due to privacy and confidentiality concerns, the original features were transformed using Principal Component Analysis (PCA), resulting in anonymized features (V1–V28) instead of the original transaction details.

In addition, the dataset was examined for missing and null values during the preprocessing stage, and no significant missing values were found.

Because of the strong class imbalance, relying solely on accuracy as a performance metric can be misleading. Therefore, more appropriate evaluation metrics such as Precision, Recall, F1-score, and Area Under the ROC Curve (AUROC) are used to assess model performance.

4.1.1 Statistical Analysis of the Dataset

The statistical analysis of selected features from the dataset is presented in Table 5 to provide an overview of data distribution.

Feature	Mean	Median	Standard Deviation	Maximum	Minimum
Time	94811.078	84692.500	47481.048	172792.000	0.000
V1	0.006	0.020	1.948	2.455	-56.408

V2	-0.004	0.064	1.647	22.058	-72.716
V3	0.002	0.180	1.509	9.383	-48.326
V4	-0.003	-0.022	1.414	16.875	-5.683
Amount	88.473	22.000	250.399	25691.160	0.000
Class	0.002	0.000	0.041	1.000	0.000

Table 5: Statistical Analysis of Selected Features

The statistical results indicate that most PCA-transformed features are centered around zero, which is expected due to the anonymization process applied to the dataset. In contrast, the Amount feature exhibits a considerably wider range and larger standard deviation, reflecting substantial variation in transaction values. The target variable (Class) has a mean value close to zero because fraudulent transactions constitute only a very small proportion of the dataset. These observations confirm the highly imbalanced nature of the dataset and highlight the importance of applying appropriate preprocessing and evaluation techniques when developing fraud detection models.

4.1.2 Correlation Analysis of the Dataset

Table 6 presents the correlation between selected features and the target variable (Class), highlighting the most relevant attributes for fraud detection.

Feature Pair	Correlation Coefficient
V11 and Class	0.149
V4 and Class	0.129
V2 and Class	0.085
V19 and Class	0.034
V8 and Class	0.033

Table 6: Correlation between selected features and the target variable

It can be observed that features such as V11, V4, and V2 have relatively higher correlation with the target variable compared to other features. However, the overall correlation values remain low, indicating that fraud detection is a complex problem that

cannot be effectively addressed using simple linear relationships alone. This suggests that more advanced machine learning techniques capable of capturing non-linear patterns may be more suitable for identifying fraudulent transactions.

4.2. Experimental Setup

The experiment was conducted using Python in the Google Colab environment, utilizing libraries such as Pandas, Scikit-learn, Imbalanced-learn, Seaborn, and Matplotlib [10].

The Credit Card Fraud Detection dataset, which contains highly imbalanced classes, was used [1]. To ensure reproducibility and stable experimental results, fixed random seeds (`random_state=42`) were applied throughout the workflow, including data splitting, SMOTE, cross-validation, and model initialization.

The overall experimental workflow is illustrated in Figure 3.

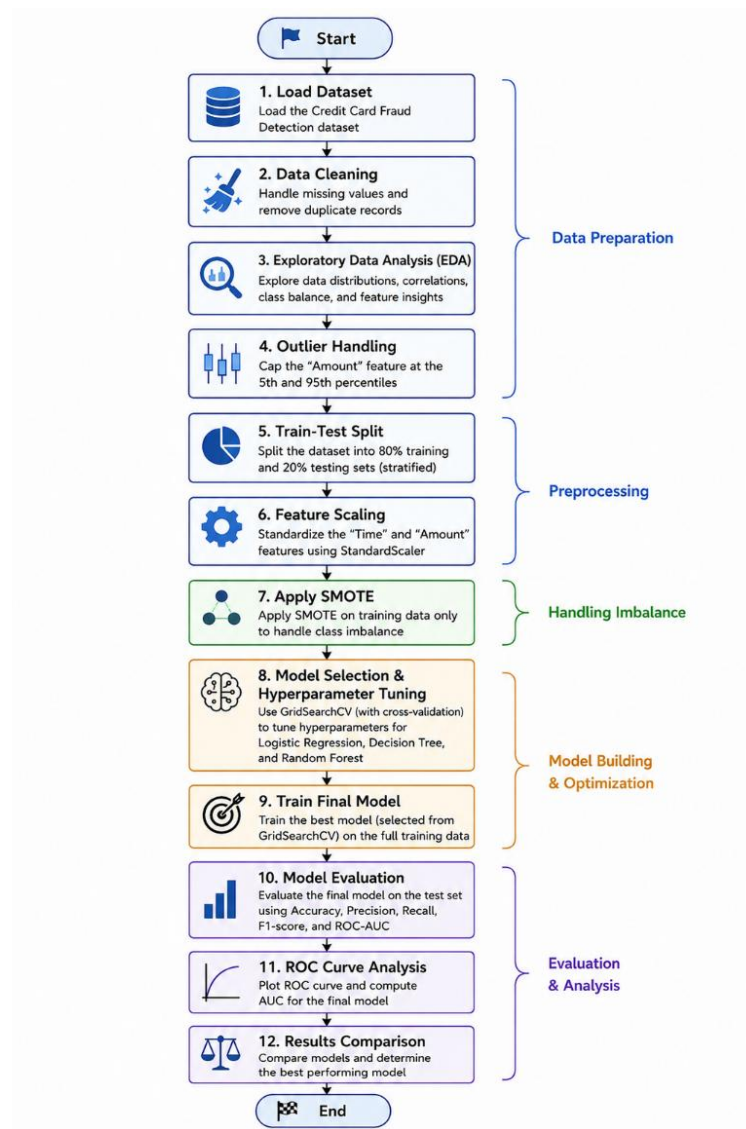


Figure 3: Flowchart of the proposed machine learning methodology for credit card fraud detection.

The process began with data preprocessing, including checking for missing values, removing duplicate records, and performing exploratory data analysis (EDA) such as class distribution analysis, descriptive statistics, and correlation analysis. Outlier treatment was then applied by capping the “Amount” feature at the 5th and 95th percentiles.

The dataset was partitioned into training and testing sets using an 80/20 split with stratified sampling to preserve the original class distribution.

Feature scaling was performed using StandardScaler to standardize numerical features, particularly the “Time” and “Amount” features. The scaler was fitted on the training data and subsequently applied to both training and testing sets.

The original dataset features (V1–V28) were already transformed using Principal Component Analysis (PCA) by the dataset providers to preserve confidentiality and protect sensitive financial information. Therefore, PCA was not re-applied for feature extraction in this project. Instead, PCA-based visualization techniques were used only for exploratory analysis and visualization purposes.

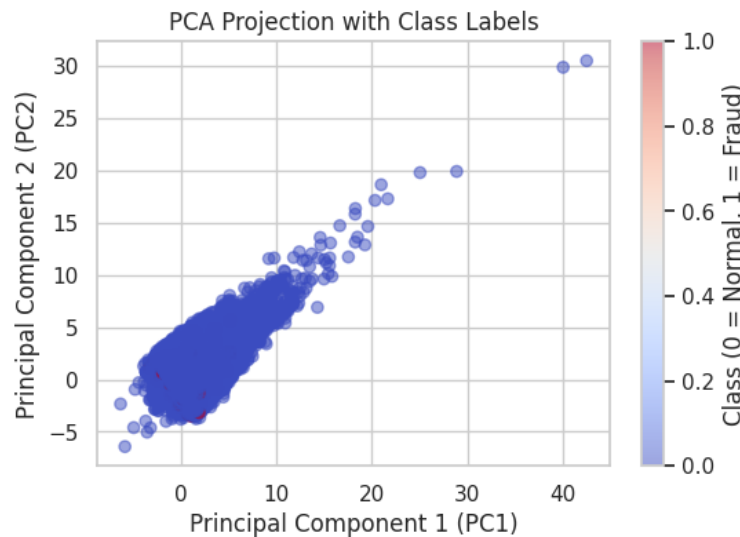


Figure 4: PCA projection of legitimate and fraudulent transactions using the first two principal components.

To address class imbalance, SMOTE (Synthetic Minority Over-sampling Technique) was applied to the training data only [2].

Three machine learning classification models were implemented: Logistic Regression, Decision Tree, and Random Forest. For the Random Forest model, `n_jobs=1` was specified to reduce variability caused by parallel processing and improve reproducibility across multiple runs.

The models were initially evaluated using Stratified K-Fold cross-validation (3-fold) on the training data.

In addition to evaluating the models using the complete feature set, a feature selection experiment was conducted to investigate the impact of reducing the number of input features on fraud detection performance. Features were ranked according to their absolute correlation with the target variable (Class). Different feature subsets, including Top 15, Top 10, and Top 5 features, were evaluated using the Random Forest classifier combined with SMOTE and Stratified K-Fold cross-validation. The purpose of this experiment was to determine whether reducing the feature space could improve computational efficiency while maintaining effective classification performance.

Finally, the best-performing model was trained on the full training dataset and evaluated on the test set using Accuracy, Precision, Recall, and F1-score. In addition, confusion matrix and ROC curve analysis were conducted to further assess classification performance and model effectiveness [3].

Parameter	Value
Programming Environment	Google Colab
Programming Language	Python
Train-Test Split	80% / 20%
Random State	42
Scaling Method	StandardScaler
Oversampling Technique	SMOTE
Cross-Validation Method	Stratified K-Fold
Number of Folds	3
Classification Models	Logistic Regression, Decision Tree, Random Forest

Table 7: Experimental Configuration

Table 7 summarizes the main experimental settings adopted in this study. These configurations were selected to ensure reproducibility, maintain class distribution during model evaluation, and provide a fair comparison among the implemented machine learning models.

4.2.1 Optimization Strategy

Hyperparameter tuning was performed using GridSearchCV with Stratified K-Fold cross-validation to identify the optimal configuration for the Random Forest classifier. A

machine learning pipeline was constructed by integrating SMOTE with the Random Forest model to ensure that class balancing was applied correctly during cross-validation and to avoid data leakage. Figure 5 illustrates the optimization workflow used in this study.

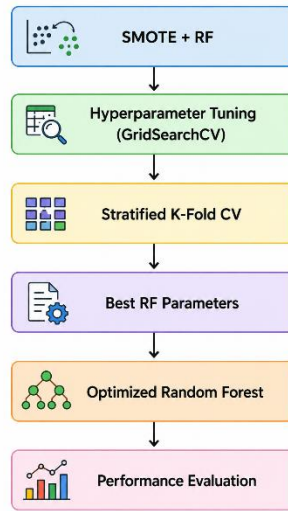


Figure 5: Hyperparameter tuning workflow using GridSearchCV and Stratified K-Fold cross-validation.

Different Random Forest configurations were evaluated using the F1-score as the primary optimization metric. The hyperparameter search focused on the maximum tree depth while keeping the number of estimators fixed at 50. The best-performing parameter combination was then selected to build the final optimized Random Forest model. Table 8 summarizes the hyperparameter tuning configuration used in this study

Parameter	Tested Values
n_estimators	50
max_depth	None, 10
Cross-Validation Method	Stratified K-Fold
Number of Folds	3
Scoring Metric	F1-score

Oversampling Technique	SMOTE
Optimization Method	GridSearchCV

Table 8: Hyperparameter tuning configuration for the Random Forest model.

Note: The hyperparameter search space was intentionally limited to a single value for $n_estimators$ (50) due to computational resource constraints in the Google Colab environment. Future work may explore a wider range (e.g., 50, 100, 200) for more thorough optimization.

Parameter	Optimal Value
$n_estimators$	50
max_depth	None

Table 9: Optimal Hyperparameters of the Random Forest Model

The optimal hyperparameter configuration identified by GridSearchCV consisted of 50 decision trees with no restriction on the maximum tree depth ($max_depth=None$). This configuration achieved the highest F1-score during cross-validation and was therefore selected to train the final Random Forest model used in the subsequent evaluation.

4.3 Performance Measures

The performance of the proposed machine learning models is evaluated using standard classification metrics, including Accuracy, Precision, Recall, and F1-score. These metrics are widely used in fraud detection problems to evaluate model effectiveness, especially when dealing with imbalanced datasets [14].

Accuracy measures the overall correctness of the model by calculating the proportion of correctly classified transactions. However, in highly imbalanced datasets such as credit card fraud detection, accuracy alone may be misleading, since the majority class (legitimate transactions) dominates the data.

Precision measures the proportion of transactions predicted as fraudulent that are actually fraudulent. A high precision value indicates a low number of false positive predictions, which is important to avoid incorrectly flagging legitimate transactions as fraud [14].

Recall measures the ability of the model to identify fraudulent transactions correctly. It is extremely important in fraud detection, as missing fraudulent transactions (false negatives) can lead to serious financial losses [14].

F1-score is the harmonic mean of precision and recall, providing a balanced assessment of the model's performance. It is especially useful when there is a trade-off between precision and recall [14].

AUROC (Area Under the Receiver Operating Characteristic Curve) measures the model's ability to distinguish between legitimate and fraudulent transactions across different classification thresholds. A higher AUROC value indicates better discrimination capability and overall classification performance, making it particularly useful for evaluating fraud detection models on imbalanced datasets [14].

In addition, the **confusion matrix** is used to provide a detailed assessment of the model's classification performance. It displays the number of true positives, true negatives, false positives, and false negatives, offering a clear understanding of how the model performs in both classes.

To ensure reliable and reproducible evaluation results, fixed random seeds ("random_state=42") were applied during data splitting, SMOTE resampling, cross-validation, and model initialization. Moreover, Stratified K-Fold cross-validation was used to preserve the original class distribution across folds during model evaluation.

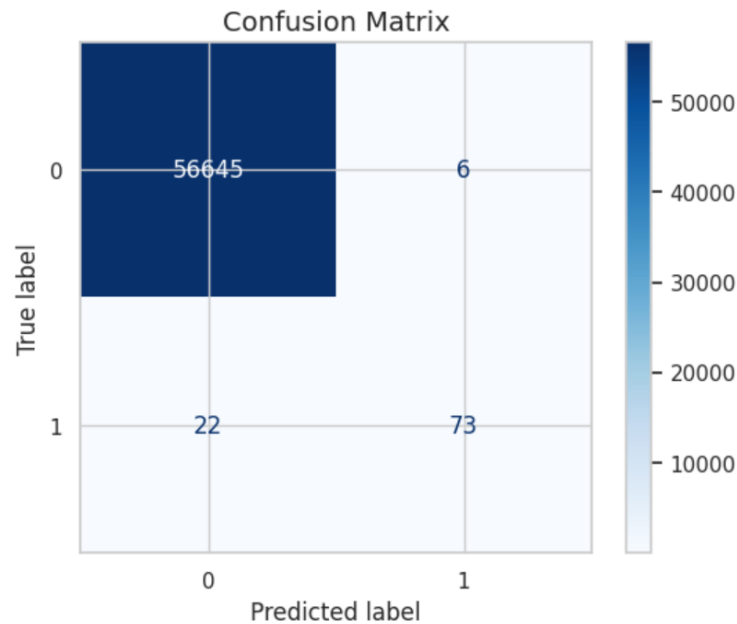


Figure 6 shows the confusion matrix for the Random Forest model, which provides further insight into the classification results and supports the evaluation of accuracy and recall.

Actual / Predicted	Legitimate (0)	Fraudulent (1)
Legitimate (0)	56645	6

Fraudulent (1)	22	73
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Table 10: Confusion Matrix Values for Random Forest Model

The confusion matrix shows that the Random Forest model correctly classified 56,645 legitimate transactions and 73 fraudulent transactions. Only six legitimate transactions were incorrectly classified as fraud, indicating a very low false positive rate. However, 22 fraudulent transactions were misclassified as legitimate, representing approximately 23% of fraud cases in the test set. This result highlights the model's strong precision while also indicating that further improvements are needed to increase recall and reduce missed fraud cases in real-world applications.

5.0 Results and Discussion

This section presents the experimental results obtained from the implemented machine learning models for credit card fraud detection. Initially, the models were evaluated using the complete feature set to identify the most effective classification technique. Subsequently, a feature selection experiment was conducted to investigate the impact of reducing the number of features on classification performance. Finally, the best-performing model was selected, analyzed in detail, and compared with previous studies that used the same dataset.

5.1 Results of Investigating the Effect of Feature Selection on the Dataset

Initially, the complete feature set was used to evaluate the performance of the three implemented machine learning models: Logistic Regression, Decision Tree, and Random Forest. The evaluation was conducted using Accuracy, Precision, Recall, and F1-score metrics. Table 11 summarizes the obtained results.

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.973	0.053	0.873	0.100
Decision Tree	0.997	0.392	0.694	0.501
Random Forest	0.999	0.924	0.768	0.839

Table 11: Results Using the Complete Feature Set

As shown in Table 11, the Random Forest model achieved the highest overall performance among all evaluated classifiers. The model obtained the highest Accuracy, Precision, and F1-score, indicating its strong capability in detecting fraudulent transactions while minimizing false positive predictions. Consequently, Random Forest was selected for further investigation during the feature selection experiment.

To study the effect of feature selection on fraud detection performance, feature subsets were initially selected based on correlation analysis. Different feature subsets were then evaluated using the Random Forest classifier combined with SMOTE and Stratified K-Fold cross-validation. The objective was to determine whether reducing the number of features could improve classification performance while maintaining effective fraud detection capability.

Feature Subset	Number of Features	Precision	Recall	F1-Score
All Features	30	0.877	0.820	0.847
Top 15 Features	15	0.851	0.812	0.831
Top 10 Features	10	0.794	0.816	0.805
Top 5 Features	5	0.494	0.808	0.613

Table 12: Results of Different Feature Subsets

In addition, feature importance analysis was performed using the Random Forest classifier to identify the most influential features contributing to fraud detection performance.

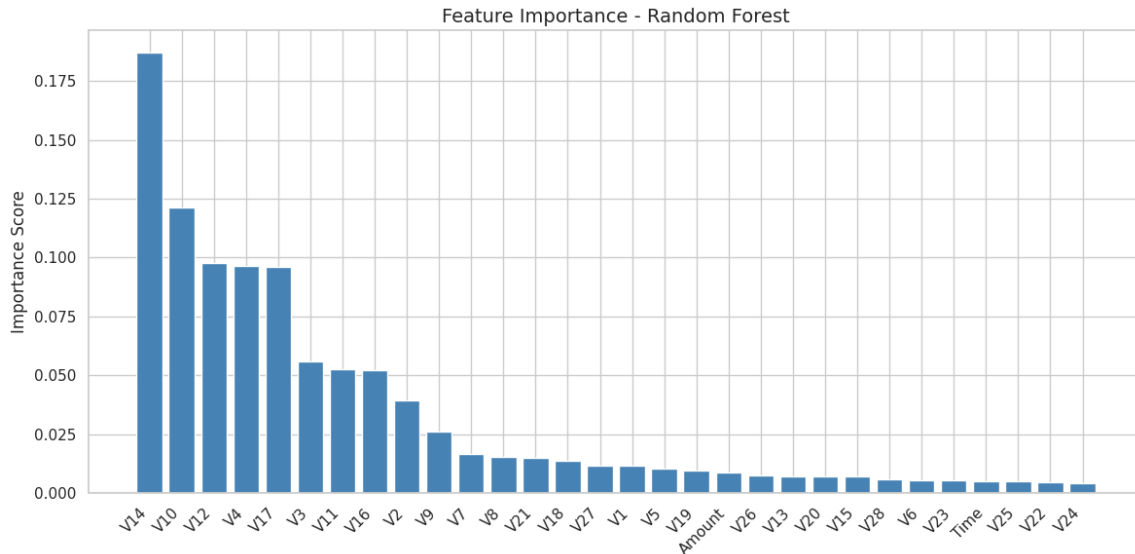


Figure 7: Feature Importance Scores Obtained from the Random Forest Model.

Figure 7 illustrates the feature importance scores generated by the Random Forest classifier. The results indicate that features such as V17, V14, V12, V10, and V4 contributed most significantly to fraud detection performance. In contrast, Time and several lower-ranked PCA components showed relatively weak correlations with the target class. These findings support the feature selection analysis and demonstrate that combining multiple informative features leads to improved classification performance.

The results indicate that the complete feature set achieved the highest overall performance, obtaining a Precision of 0.877, Recall of 0.820, and F1-score of 0.847. As the number of selected features decreased, classification performance gradually deteriorated. The most significant reduction occurred when only the top five features were retained, resulting in a substantial decrease in overall classification performance, particularly in Precision and F1-score.

These findings suggest that although certain features exhibit stronger correlations with the target variable, the remaining features still contribute valuable complementary information for fraud detection. Therefore, retaining all available features provided the most effective classification performance and was selected for the final model development.

5.2 Discussion of Final Results

Based on the experimental results presented in Section 5.1, the Random Forest classifier using the complete feature set was selected as the final model. The model was further optimized using GridSearchCV with Stratified K-Fold cross-validation to identify the best hyperparameter configuration.

Metric	Value
Accuracy	99.95%
Precision	92.41%
Recall	76.84%
F1-Score	83.91%
AUROC	0.9519

Table 13: Results of the Final Optimized Random Forest Model

The final optimized Random Forest model achieved excellent classification performance, obtaining an accuracy of 99.95%, precision of 92.41%, recall of 76.84%, and an F1-score of 83.91%. These results indicate that the model successfully identified the majority of fraudulent transactions while maintaining a very low false positive rate.

The confusion matrix results presented earlier in Table 10 show that the model correctly classified 56,645 legitimate transactions and 73 fraudulent transactions, while only six legitimate transactions were incorrectly classified as fraudulent. However, 22 fraudulent transactions were misclassified as legitimate transactions. These results demonstrate the model's **strong precision** while highlighting opportunities for further improving **recall**.

The recall value of 76.84% indicates that approximately 23% of fraudulent transactions remained undetected. Although the model effectively minimizes false positive alerts, these missed fraud cases may still result in financial losses in real-world financial environments. Therefore, future improvements should focus on increasing recall while preserving the model's strong precision and low false positive rate.

Figure 8 illustrates the ROC curve of the optimized Random Forest classifier.

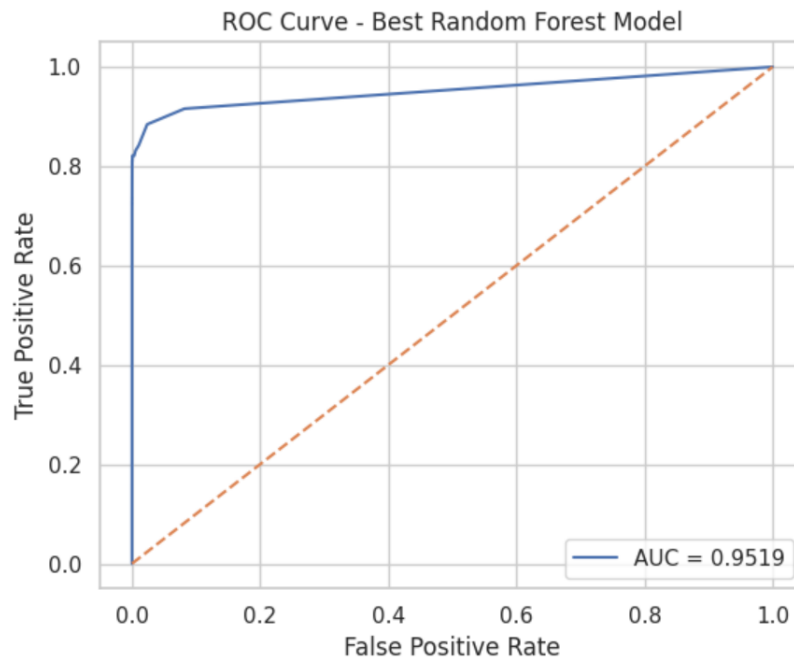


Figure 8: ROC Curve of the Final Random Forest Model.

The Random Forest model achieved an AUROC value of 0.9519, indicating strong discrimination capability between fraudulent and legitimate transactions across different classification thresholds. This result confirms the robustness and reliability of the proposed fraud detection system.

Furthermore, Logistic Regression achieved the highest recall among the evaluated models. However, its very low precision indicates that many legitimate transactions were incorrectly classified as fraudulent. Although such behavior may improve fraud detection sensitivity, it would generate an excessive number of false alarms in practical

applications. In contrast, Random Forest provided the most balanced trade-off between fraud detection capability and operational efficiency.

5.3 Further Discussions

To further evaluate the effectiveness of the proposed approach, the obtained results were compared with selected previous studies that utilized the same Credit Card Fraud Detection dataset from Kaggle.

Study	Dataset	Algorithms Used	Best Performance
Niu et al. (2019)	Kaggle Credit Card Fraud Dataset	LR, KNN, SVM, DT, RF, XGB	RF AUROC = 0.988
Proposed Work	Kaggle Credit Card Fraud Dataset	LR, DT, RF	RF AUROC = 0.9519

Table 14: Comparison with Previous Studies

The comparison demonstrates that previous studies achieved strong fraud detection performance using sophisticated ensemble and hybrid learning techniques. For example, Niu et al. reported an AUROC of 0.988 using ensemble-based machine learning models.

Although the proposed Random Forest model achieved a slightly lower AUROC value of 0.9519, it provided highly competitive fraud detection performance while maintaining a simpler implementation framework. The use of data preprocessing, feature scaling, SMOTE oversampling, and hyperparameter optimization contributed significantly to improving classification performance and reducing the impact of class imbalance.

The obtained results confirm that Random Forest remains a highly effective solution for credit card fraud detection and can achieve reliable performance without requiring excessively complex modeling architectures.

5.4 Alignment with the Requirements

The proposed AI-based credit card fraud detection system successfully aligns with the requirements of financial institutions by providing an intelligent solution for detecting fraudulent transactions. The system analyzes transaction data automatically and classifies transactions as either legitimate or fraudulent using machine learning techniques.

The project addresses the challenge of highly imbalanced fraud datasets through the use of SMOTE, which improves the model's ability to identify fraudulent activities. In addition, multiple machine learning algorithms were evaluated to ensure the selection of the most effective model.

The final Random Forest model achieved high accuracy and strong fraud detection performance while maintaining a low false positive rate. This helps reduce financial losses, improve transaction security, and support faster decision-making in fraud monitoring systems.

Furthermore, the proposed solution is scalable, data-driven, and capable of supporting real-world fraud detection environments, making it suitable for the needs of modern financial organizations.

6.0 Conclusion & Recommendation

This project presented a machine learning-based approach for detecting credit card fraud using three classification algorithms: Logistic Regression, Decision Tree, and Random Forest. The experimental results demonstrated that machine learning techniques can effectively identify fraudulent transactions, even when dealing with highly imbalanced datasets. Data preprocessing techniques, including feature scaling and SMOTE oversampling, contributed to improving the classification performance of the models.

Among the evaluated models, Random Forest achieved the best overall performance with 99.95% accuracy, 92.41% precision, 76.84% recall, and an F1-score of 83.91%, making it the most reliable model for fraud detection in this study. In comparison, Logistic Regression achieved high recall but suffered from very low precision, while Decision Tree showed moderate performance compared to Random Forest.

The findings of this project confirm that ensemble learning techniques such as Random Forest can provide robust and reliable performance for fraud detection problems. In addition, the use of cross-validation and hyperparameter tuning helped improve model stability and reliability.

Despite the promising results, several limitations and areas for future improvement remain. Notably, the Random Forest model achieved a recall of 76.84%, meaning approximately 23% of actual fraudulent transactions were not detected. In a real-world financial context, these undetected fraud cases represent a meaningful risk, and improving recall without significantly sacrificing precision should be a priority in future work.

Future work may explore more advanced machine learning and deep learning techniques, such as Artificial Neural Networks and Long Short-Term Memory (LSTM) models, to further enhance fraud detection capability. Additionally, applying more advanced data balancing methods and real-time fraud detection systems may improve practical applicability in real-world financial environments. Finally, using larger and more diverse datasets could improve model generalization and robustness.

Acknowledgement

We would like to express our sincere gratitude to our supervisor, Dr. Sarah Alyami, for the valuable guidance, support, and feedback provided throughout this project.

We also thank the Department of Computer Science at Imam Abdulrahman Bin Faisal University for providing the academic environment and resources necessary for completing this work.

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